

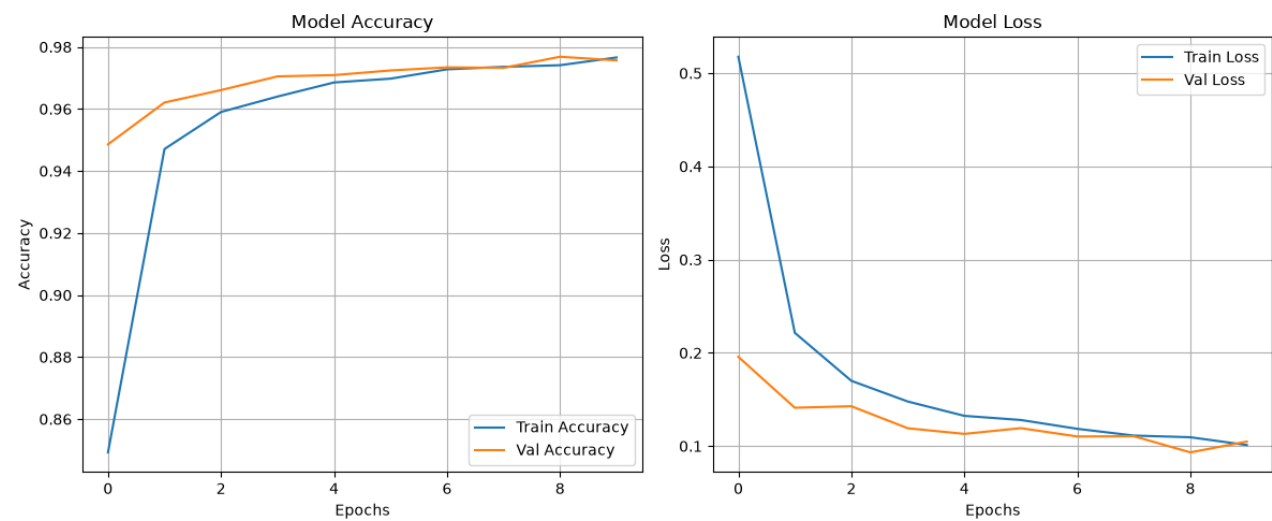
Summary

- **Baseline architecture (128→64, ReLU):** 0.9718 test accuracy
- **Activation function comparison** (same 128→64 architecture, 7 activations tested): **ELU won** (0.9777), narrowly beating LeakyReLU/Tanh (~0.975), with plain ReLU finishing *last* (0.9718) among all variants tested. Sigmoid and SELU lagged.
- **Architecture search** (width/depth, using ELU/GELU): Widening and deepening both helped. Best result: **512→256→128→64→32, all GELU** — 0.9808 test accuracy, with the smoothest, still-improving validation curve.
- **Dropout regularization** (same 5-layer GELU architecture, rates 0.2–0.5): No dropout rate improved on the no-dropout baseline except **0.2**, which tied it exactly (0.9808). Higher rates (0.3→0.5) showed a smooth accuracy decline
- **L2 regularization** (weight_decay=1e-4, layered on top of each dropout rate): Dropout+L2 combination scored below its dropout-only counterpart. Though, dropout=0.3+L2 was the *smoothest*-fitted even though not the highest-scoring, while dropout=0.5+L2 did not converge
- **Early stopping** (dropout=0.3, L2=1e-4, patience=3): Training ran the full 10 epochs, patience counter reset every time it hit 1 (never reached the threshold of 3), meaning validation loss kept finding new minima just often enough to avoid triggering a stop. The lowest validation loss occurred at **epoch 9** (0.0933).

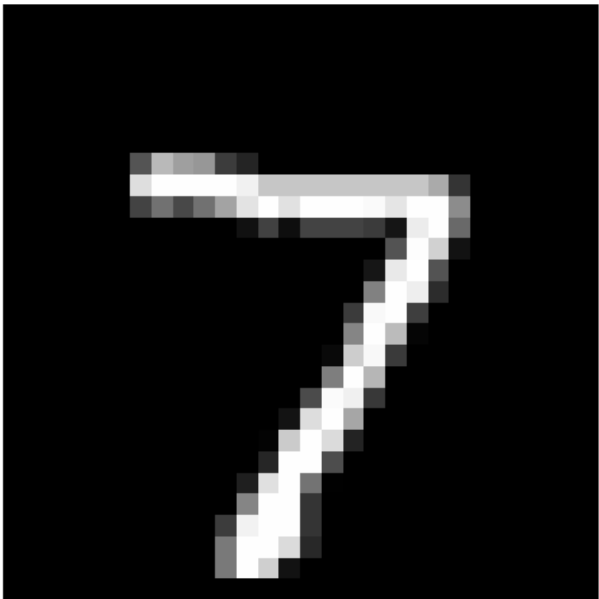
Final Model:

- Batch Size: 32
- Hidden Layers: (512, 256, 128, 64, 32)
- Activation: All GELU
- Drop-Out: 0.3
- L2 Reg: 1e-4
- Optimizer: Adam
- Test Accuracy: 0.9803

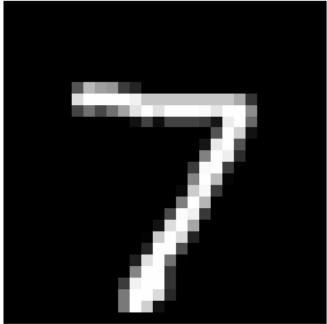
Umair Jamal (538263)



True Label: 7, Predicted: 7



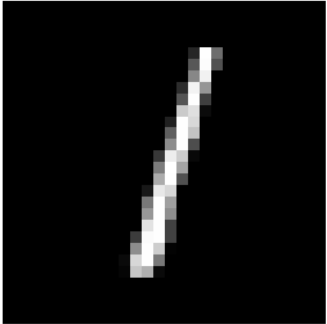
True: 7, Predicted: 7



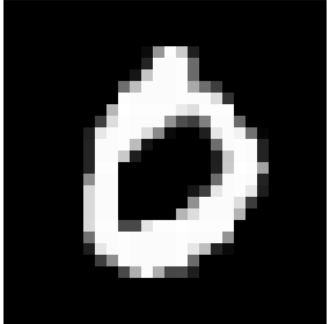
True: 2, Predicted: 2



True: 1, Predicted: 1



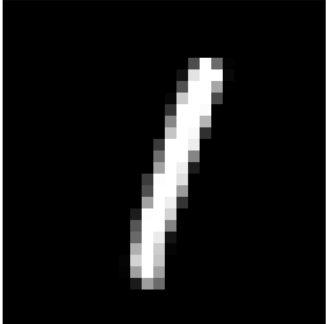
True: 0, Predicted: 0



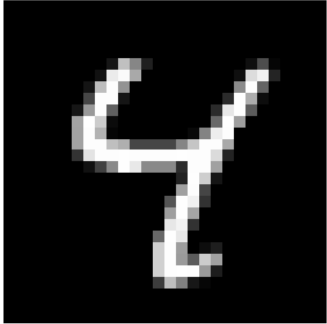
True: 4, Predicted: 4



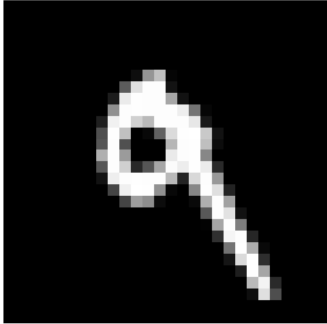
True: 1, Predicted: 1



True: 4, Predicted: 4



True: 9, Predicted: 9



True: 5, Predicted: 6

